

MECHANICAL FAILURE INVESTIGATION REPORT

Aspen Forensic Mechanical · Our reference AFM-2026-0211

INSTRUCTIONS

Instructed by:	Front Range Commercial Insurance Group, Inc.
On behalf of:	Beacon Mechanical Services, Inc.
Policy number:	
Author:	Lucien Okonkwo-Hartley, PE
Instructed by:	Front Range Mutual Casualty Insurance Company
Instructed on:	9 February 2026
Site attendance:	10 February 2026
Scope:	Cause of the coupling separation and the allocation of loss

PROPERTY AFFECTED, BY OWNER

Suite 410 — third party:	Front Range Orthopaedic, ceiling and flooring
Suite 415 — third party:	Cherry Creek Dermatology, ceiling and flooring
Suite 315 — third party:	Cherry Creek Dental Arts, panoramic X-ray unit
Level 1 mechanical room:	Insured's installation materials, 400 ft copper
Level 1 mechanical room:	Insured's tools — threading machine, crimpers

MECHANICAL FAILURE INVESTIGATION REPORT

Circumstances and findings

CIRCUMSTANCES

Circumstances:

The insured is the mechanical subcontractor on a chiller plant replacement and hydronic distribution upgrade in an occupied and operating medical office building. Its scope is the rooftop plant, the Level 1 mechanical room and the risers. Riser CHW-3 was filled and pressurised on 7 February.

FINDINGS

Findings:

The separated coupling is a 6-inch rigid grooved coupling. The gasket is rolled over approximately 90 degrees of its circumference and shows extrusion into the bolt pad gap. Bolt torque measured on the retained housing halves is 61 and 68 N·m against a specified 136 N·m. The joint held during the fill and separated under overnight thermal and pressure cycling, which is consistent with an under-torqued joint and a rolled gasket at assembly rather than with a manufacturing defect. The joint was assembled by the insured's own foreman. The coupling, gasket, bolts and pipe ends are retained.

Date of loss:	7 February 2026
Cause of loss:	Escape of water — grooved coupling separation on a newly installed riser
Incident reference:	BMS-2026-0207
Injuries reported:	0

MECHANICAL FAILURE INVESTIGATION REPORT

Preliminary quantum and reserve advice

PRELIMINARY QUANTUM

Basis:

The figures below are preliminary and are given for reserving purposes only. They are subject to verification and may move as the investigation continues.

Third-party building reinstatement, Level 1	USD 240,000
Third-party tenant contents and equipment	USD 185,000
Insured's own installation materials:	USD 46,000
Insured's own tools and equipment:	USD 38,000
Riser and coupling rework:	USD 22,000
Mitigation billed to the building owner:	USD 60,000
Estimated loss:	590,000
Currency:	USD

ON WHICH POLICY ANSWERS, AND THE TESTING CONDITION

Comment:

Roughly four fifths of the loss is third-party damage to an occupied building and its tenants, which is a liability exposure. The insured's own installation materials, its tools and the rework of the defective joint are first-party and belong on its installation and equipment floater, on which the cost of making good the defective coupling itself is in any event excluded. Both policies carry a mirrored water damage control condition requiring that systems are not left charged outside working hours unless the supply is isolated or leak detection with automatic shut-off is operational. The system was left charged over a weekend with nobody on site; whether either exception was satisfied has not yet been established and materially changes the retention.

RECOMMENDATION

Recommendation:

Set the primary file under the liability policy and open a companion file on the installation floater. Obtain the fill and pressure test record and the written notice to the general contractor before drawing any conclusion on the condition.